

Deepanjay Nandal

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PROFESSIONAL SUMMARY

Backend Software Engineer focused on distributed backend and AI infrastructure, with 2.5+ years of engineering experience spanning Android/AOSP validation, test automation, and systems debugging. M.S. in Software Engineering from Arizona State University, with work in Kafka/Redis pipelines, hybrid RAG/vector-search systems, and LLM inference caching.

EDUCATION

Arizona State University
Master of Science (M.S.) in Computer Software Engineering
Vellore Institute of Technology
Bachelor of Technology (B.Tech) in Information Technology

Tempe, Arizona, USA
Aug 2024 – May 2026
Tamil Nadu, India
Jul 2017 – Jul 2021

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL
Backend Technologies: Distributed Systems, Microservices, Event-Driven Architecture, Multi-tenancy, REST APIs
Data & Infrastructure: Apache Kafka, PostgreSQL, Redis, pgvector, Distributed Caching, BullMQ
Security & Systems: Android Open Source Project (AOSP), Vulnerability Research, Binary Analysis, Reverse Engineering, Ghidra, Frida, ClusterFuzz
Cloud & DevOps: Amazon Web Services (AWS), Amazon EC2, AWS Lambda, Amazon SQS, AWS IoT Greengrass, Serverless Architecture, Edge Computing, Docker, CI/CD
AI/ML: Retrieval-Augmented Generation (RAG), Vector Search, Large Language Model (LLM) APIs, Semantic Caching, Embeddings, Retrieval Evaluation

EXPERIENCE

Discover Excellence LLC
Software Engineering Intern

Remote (California, USA)
Feb 2026 – May 2026

- Built a Kafka-based visitor intelligence pipeline for an AI-powered eCommerce analytics platform, processing 100k+ daily events with sub-40ms p95 latency and improving reporting observability.
- Designed a multi-tenant resource manager using Redis locks, preventing worker contention and enforcing tenant isolation and fair resource usage across client workloads.
- Implemented reliable background processing with BullMQ, retries, exponential backoff, and idempotency keys, achieving 99.9% event-processing reliability.

Ittiam Systems Pvt Ltd
Software Engineer / Senior Software Engineer

Bangalore, India
Mar 2022 – Jun 2024

- Built modular Android/AOSP validation infrastructure across Bluetooth, Media, and core platform modules, integrating 40+ reusable CTS/STS suites to automate vulnerability reproduction and regression coverage.
- Led Ghidra-based binary and assembly analysis across framework, native, and OEM partner builds, validating security patch coverage for 50+ CVEs and strengthening regression confidence.
- Engineered Python automation pipelines to process ClusterFuzz crash outputs, orchestrate Frida-based runtime validation, and standardize CVE triage workflows, reducing manual investigation time by 40%.
- Mentored 4 junior engineers on CTS/STS test authoring, crash triage, code reviews, and security validation workflows, improving onboarding quality and review consistency.

PROJECTS

BitMod (AI Inference Cache & Semantic Reuse Engine) | *Python, FastAPI, Embeddings, LLM Pipelines*

- Engineered a 9-layer LLM cache proxy with SHA-256 exact lookup, semantic reuse, fuzzy matching, and Bayesian evidence scoring, achieving 71ms average cached-response latency and reducing LLM API calls by up to 94%.
- Implemented pluggable provider, storage, and custom VectorIndex layers with TTL/LRU eviction, using threshold-tuned semantic matching and negative evidence to control stale or unsafe cache hits.

CaseTally (AI-Powered Legal Search Platform) | *FastAPI, pgvector, RAG, BM25, LLM APIs*

- Developed a hybrid RAG system combining BM25 and HNSW-indexed pgvector search across 33k+ U.S. Code chunks, with LLM-based query rewriting, min-max score normalization, and weighted fusion.
- Built idempotent version-hashed ingestion pipeline and FastAPI SSE streaming backend; validated retrieval quality with Precision@3 and MRR, and measured latency at p50 <35ms and p95 <100ms via an evaluation harness.

Elastic Face Recognition Cloud System | *AWS, EC2, SQS, Lambda, Docker, IoT Greengrass*

- Scaled face-recognition inference using MTCNN and FaceNet in PyTorch across 15 EC2 instances with an SQS queue-depth autoscaler, achieving 0.96s latency under 100 concurrent requests with 100% classification accuracy.
- Re-architected pipeline into serverless and edge-cloud variants using AWS Lambda images, Docker, ECR, AWS IoT Greengrass, and MQTT, achieving 776ms pipeline response time by moving MTCNN detection to edge devices.